Power Plants and Quality of Life in Southeastern US: Final Script

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Setup

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.2     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(readxl)
library(here)
```

```
## here() starts at /home/guest/ENV710/ENV710_GroupProj
```

```r
library(tidycensus)
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```r
library(glue)
library(moments)
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
library(openxlsx)
library(ggplot2)
library(moments)
library(stats)
library(nlme)
```

```
##
## Attaching package: 'nlme'
##
## The following object is masked from 'package:dplyr':
##
##     collapse
```

```r
library(naniar)
library (DHARMa)
```

```
## Warning in check_dep_version(): ABI version mismatch:
## lme4 was built with Matrix ABI version 2
## Current Matrix ABI version is 1
## Please re-install lme4 from source or restore original 'Matrix' package

## This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type news(package = 'DHARMa')
```

```r
library(gtsummary)
library(broom)
library(sf)
```

```
## Linking to GEOS 3.10.2, GDAL 3.4.1, PROJ 8.2.1; sf_use_s2() is TRUE
```

```r
library(corrplot)
```

```
## corrplot 0.94 loaded
```

Data Wrangling

CVI

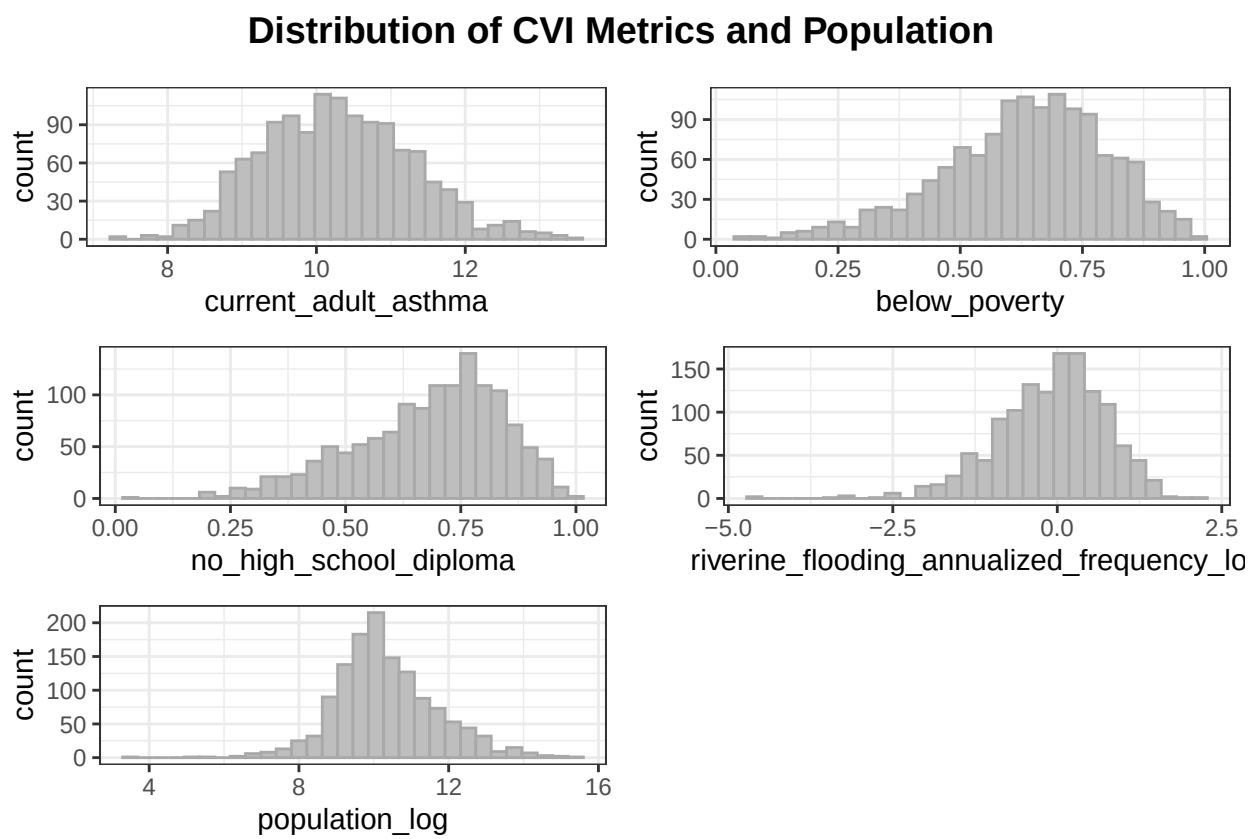
```r
southern_states <- c("FL", "GA", "AL", "MS", "LA", "AR", "TN", "NC", "SC",
                     "KY", "WV", "VA", "TX")

columnlist <- c("fips_code",
                "state",
                "current_adult_asthma",
                "below_poverty",
                "no_high_school_diploma",
                "riverine_flooding_annualized_frequency")
```

```r
cvi_data <- read_excel(path=here('Data','Raw','cvi',"CVI_subset.xlsx"),
                       skip = 0,
                       sheet="Indicator Native Units",
                       col_names=T) %>%
  clean_names() %>%
  filter(state %in% southern_states) %>%
  select(all_of(columnlist))


# Get population at the census tract and county level --------------------------------

# census-tract population for population-weighted averages
census_tract_pop <- get_acs(geography = "tract",
                            state = southern_states,
                            variables = "B01003_001",
                            geometry = FALSE) %>%
  select("fips_code" = GEOID, "population" = estimate)

# county population to use for analysis
county_pop <- get_acs(geography = "county",
                      state = southern_states,
                      variables = "B01003_001",
                      geometry = FALSE) %>%
  select("fips_code" = GEOID, "county_name" = NAME, "population" = estimate)

# Join population and clean CVI data ---------------------------------------

# ASSUMPTION: for tracts that do not have population, assign a population of 1

cvi_county <- cvi_data %>%
  left_join(census_tract_pop, by = "fips_code") %>%
  mutate(population = replace_na(population, 1)) %>%
  mutate(fips_county = str_extract(fips_code,"^\\d{5}")) %>%
  group_by(state, fips_county) %>%
  # summarize tract data to weighted average, weighted by population
  summarize(across(c("current_adult_asthma":"riverine_flooding_annualized_frequency"),
                   ~sum(population*., na.rm = TRUE)/sum(population, na.rm = TRUE))) %>%
  ungroup() %>%
  # add on county-level population data
  # exclude counties without population data in the census
  inner_join(county_pop, by = c("fips_county" = "fips_code"))


# Investigate Variable Distribution ----------------------------------------

metrics <- cvi_county %>% select(-c(state:fips_county, county_name)) %>% names()

# visualize distributions
plot_distrib <- function(metric, county_df, include_skew=TRUE){
  df <- county_df %>%
    select(fips_county, all_of(metric))

  metric_data <- df %>% select(all_of(metric)) %>% pull()
```

```r
  skewness <- skewness(metric_data %>% subset(.!=-Inf), na.rm = TRUE)
  kurtosis <- kurtosis(metric_data %>% subset(.!=-Inf), na.rm = TRUE)

  plot <- ggplot(df, aes(x = !!sym(metric))) +
    geom_histogram(fill = "grey", color = "darkgrey") +
    theme_bw()

  if(include_skew){
    plot <- plot + labs(caption = glue("Skewness: {skewness}\nKurtosis: {kurtosis}"))
  }
  return(list(plot, skewness))
}

plots <- map(metrics, plot_distrib, cvi_county)
names(plots) <- metrics

#plots

# Log Adjust Variables -----------------------------------------------------

# log adjust across all cvi metrics
cvi_county_log_adj <- cvi_county %>%
  select(fips_county, all_of(metrics)) %>%
  mutate(across(c(-fips_county), log))

plots_log <- map(metrics, plot_distrib, cvi_county_log_adj)
names(plots_log) <- metrics
#plots_log

cvi_county_final <- left_join(
  cvi_county,
  cvi_county_log_adj,
  by = "fips_county",
  suffix = c("", "_log")
) %>%
  relocate(county_name, .after = "fips_county")


# check whether original or log-transformed variable is better
best_col <- c()
for (metric in metrics){
  orig_skew <- plots[metric][[1]][[2]]
  log_skew <- plots_log[metric][[1]][[2]]

  if((abs(orig_skew) < abs(log_skew)) | (abs(orig_skew)<0.5)){
    best_col <- best_col %>% append(metric)
  }
  else{
    best_col <- best_col %>% append(str_c(metric, "log", sep = "_"))
  }
}
best_col
```

```
## [1] "current_adult_asthma"
## [2] "below_poverty"
## [3] "no_high_school_diploma"
## [4] "riverine_flooding_annualized_frequency_log"
## [5] "population_log"
```

```r
cvi_county_best_cols <- cvi_county_final %>%
  select(state:county_name, all_of(best_col))

#write_rds(cvi_county_final, "Data/Processed/cvi_data_by_county.rds")
#write_rds(cvi_county_best_cols, "Data/Processed/cvi_data_by_county_best_cols.rds")

# Visualize variable distributions -----------------------------------------------------

# plot grid of metrics
plots_best_cols <- map(best_col, plot_distrib, cvi_county_best_cols, FALSE)
plots_best_cols <- lapply(plots_best_cols, function(x) x[[1]])  # extract just the plot

title <- ggdraw() +
  draw_label("Distribution of CVI Metrics and Population",
             fontface = 'bold', size = 14, hjust = 0.5)

final_distributions <- plot_grid(
  title,
  plot_grid(plotlist = plots_best_cols, nrow = 3),
  nrow = 2, rel_heights = c(0.1, 0.9)
) +
  theme(
    plot.background = element_rect(fill = "white", color = NA),
    panel.background = element_rect(fill = "white", color = NA)
  )

final_distributions
```




```
#ggsave("Figures/Presentation/variable_distributions.png", final_distributions,
       #width = 7, height = 5, scale = 1.15)
```

eGRID

```
plnt23_raw <- read_excel(here("Data/Raw/egrid/egrid2023_data_rev2.xlsx"),
                         sheet = "PLNT23", skip=1)

glimpse(plnt23_raw)
```

```
## Rows: 12,612
## Columns: 150
## $ SEQPLT23  <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 1~
## $ YEAR      <dbl> 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, ~
## $ PSTATABB  <chr> "AK", "AK", "AK", "AK", "AK", "AK", "AK", "AK", "AK", "AK", ~
## $ PNAME     <chr> "Alakanuk", "Allison Creek Hydro", "Ambler", "Angoon", "Ania~
## $ ORISPL    <dbl> 57053, 58982, 60243, 7462, 7182, 62, 7250, 79, 7173, 57583, ~
## $ OPRNAME   <chr> "Alaska Village Elec Coop, Inc", "Copper Valley Elec Assn, I~
## $ OPRCODE   <dbl> 221, 4329, 221, 18963, 4959, 213, 213, 7353, 1276, 7353, 102~
## $ UTLSRVNM  <chr> "Alaska Village Elec Coop, Inc", "Copper Valley Elec Assn, I~
## $ UTLSRVID  <dbl> 221, 4329, 221, 18963, 4959, 213, 213, 986, 1276, 7353, 1021~
## $ SECTOR    <chr> "Electric Utility", "Electric Utility", "Electric Utility", ~
## $ BANAME    <chr> "No balancing authority - AK", "No balancing authority - AK"~
```

```
## $ BACODE    <chr> "NA - AK", "NA - AK", "NA - AK", "NA - AK", "NA - AK", "NA -~
## $ NERC      <chr> "AK", "AK", "AK", "AK", "AK", "AK", "AK", "AK", "AK", "AK", ~
## $ SUBRGN    <chr> "AKMS", "AKMS", "AKMS", "AKMS", "AKMS", "AKMS", "AKMS", "AKG~
## $ SRNAME    <chr> "ASCC Miscellaneous", "ASCC Miscellaneous", "ASCC Miscellane~
## $ ISORTO    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ FIPSST    <chr> "02", "02", "02", "02", "02", "02", "02", "02", "02", "02", ~
## $ FIPSCNTY  <chr> NA, "261", "188", NA, "050", "110", "110", "090", "185", "09~
## $ CNTYNAME  <chr> "Kusilvak", "Valdez Cordova", "Northwest Arctic", "Hoonah-An~
## $ LAT       <dbl> 62.68330, 61.08444, 67.08798, 57.49917, 61.58068, 58.31760, ~
## $ LON       <dbl> -164.6544, -146.3533, -157.8567, -134.5861, -159.5356, -134.~
## $ CAPDFLAG  <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ NUMUNT    <dbl> 1, 1, 3, 3, 3, 2, 3, 4, 5, 1, 3, 6, 3, 6, 1, 3, 2, 3, 1, 1, ~
## $ NUMGEN    <dbl> 1, 1, 3, 3, 3, 2, 3, 3, 5, 1, 3, 6, 3, 6, 1, 3, 2, 3, 1, 1, ~
## $ PLPRMFL   <chr> "DFO", "WAT", "DFO", "DFO", "DFO", "WAT", "DFO", "SUB", "NG"~
## $ PLFUELCT  <chr> "OIL", "HYDRO", "OIL", "OIL", "OIL", "HYDRO", "OIL", "COAL",~
## $ COALFLAG  <chr> NA, NA, NA, NA, NA, NA, NA, "Yes", NA, NA, NA, NA, NA, NA, N~
## $ CAPFAC    <dbl> 0.00000, 0.36812, 0.12858, 0.13820, 0.13544, 0.23924, 0.0000~
## $ NAMEPCAP  <dbl> 0.8, 6.5, 1.1, 1.5, 1.7, 4.0, 36.2, 27.5, 17.3, 40.0, 5.4, 3~
## $ NBFACTOR  <dbl> NA, NA, 1.0000000, 1.0000000, 1.0000000, NA, NA, 0.3414972, ~
## $ RMBMFLAG  <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ CHPFLAG   <chr> "Yes", NA, NA, NA, NA, NA, NA, "Yes", NA, NA, NA, NA, NA, NA~
## $ USETHRMO  <dbl> 0.0, NA, NA, NA, NA, NA, NA, 708380.8, NA, NA, NA, NA, NA, N~
## $ PWRTOHT   <dbl> NA, NA, NA, NA, NA, NA, NA, 0.691, NA, NA, NA, NA, NA, NA, N~
## $ ELCALLOC  <dbl> 1.000000, NA, NA, NA, NA, NA, NA, 0.479424, NA, NA, NA, NA, ~
## $ PSFLAG    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ PLHTIAN   <dbl> 0, NA, 13779, 23410, 26201, NA, 4738, 1288382, 773377, NA, N~
## $ PLHTIOZ   <dbl> 0.000, NA, 5218.001, 8862.999, 9920.999, NA, 1791.000, 43059~
## $ PLHTIANT  <dbl> NA, 71938, 13779, 23410, 26201, 28859, 4738, 1288382, 773377~
## $ PLHTIOZT  <dbl> NA, 61081.000, 5218.001, 8862.999, 9920.999, 0.000, 1791.000~
## $ PLNGENAN  <dbl> 0.000, 20961.000, 1238.999, 1815.999, 2016.999, 8383.000, -2~
## $ PLNGENOZ  <dbl> 0.000, 17902.000, 469.318, 687.879, 764.017, -48.000, -89.39~
## $ PLNGENNB  <dbl> NA, NA, 1238.999, 1815.999, 2016.999, NA, NA, 48957.033, 382~
## $ PLNOXAN   <dbl> NA, NA, 22.889, 30.225, 41.920, NA, 2.385, 225.467, 123.742,~
## $ PLNOXOZ   <dbl> NA, NA, 8.668, 11.445, 15.872, NA, 0.903, 70.021, 58.418, NA~
## $ PLSO2AN   <dbl> NA, NA, 1.998, 3.394, 3.799, NA, 1.174, 93.618, 1.236, NA, N~
## $ PLCO2AN   <dbl> NA, NA, 1125.234, 1911.729, 2139.649, NA, 386.918, 137787.53~
## $ PLCH4AN   <dbl> 0.000, NA, 91.133, 154.831, 173.290, NA, 31.337, 31244.406, ~
## $ PLN2OAN   <dbl> 0.000, NA, 18.227, 30.966, 34.658, NA, 6.267, 4544.641, 170.~
## $ PLCO2EQA  <dbl> 0.000, NA, 1128.925, 1918.000, 2146.667, NA, 388.187, 138827~
## $ PLHGAN    <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ PLNOXRTA  <dbl> NA, NA, 36.948, 33.287, 41.567, NA, NA, 3.145, 5.304, NA, NA~
## $ PLNOXRTO  <dbl> NA, NA, 36.939, 33.276, 41.549, NA, NA, 2.764, 5.304, NA, NA~
## $ PLSO2RTA  <dbl> NA, NA, 3.225, 3.738, 3.767, NA, NA, 1.306, 0.053, NA, NA, 0~
## $ PLCO2RTA  <dbl> NA, NA, 1816.360, 2105.430, 2121.616, NA, NA, 1922.259, 1937~
## $ PLCH4RTA  <dbl> NA, NA, 0.074, 0.085, 0.086, NA, NA, 0.218, 0.037, NA, NA, 0~
## $ PLN2ORTA  <dbl> NA, NA, 0.015, 0.017, 0.017, NA, NA, 0.032, 0.004, NA, NA, 0~
## $ PLC2ERTA  <dbl> NA, NA, 1822.318, 2112.336, 2128.575, NA, NA, 1936.762, 1939~
## $ PLHGRTA   <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ PLNOXRA   <dbl> NA, NA, 3.322, 2.582, 3.200, NA, 1.007, 0.350, 0.320, NA, NA~
## $ PLNOXRO   <dbl> NA, NA, 3.322, 2.583, 3.200, NA, 1.008, 0.325, 0.320, NA, NA~
## $ PLSO2RA   <dbl> NA, NA, 0.290, 0.290, 0.290, NA, 0.495, 0.145, 0.003, NA, NA~
## $ PLCO2RA   <dbl> NA, NA, 163.326, 163.326, 163.326, NA, 163.325, 213.892, 116~
## $ PLCH4RA   <dbl> NA, NA, 0.007, 0.007, 0.007, NA, 0.007, 0.024, 0.002, NA, NA~
## $ PLN2ORA   <dbl> NA, NA, 0.001, 0.001, 0.001, NA, 0.001, 0.004, 0.000, NA, NA~
```

```
## $ PLC2ERA   <dbl> NA, NA, 163.862, 163.862, 163.861, NA, 163.861, 215.506, 117~
## $ PLHGRA    <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ PLNOXCRT  <dbl> NA, NA, 36.948, 33.287, 41.567, NA, NA, 3.145, 5.304, NA, NA~
## $ PLNOXCRO  <dbl> NA, NA, 36.939, 33.276, 41.549, NA, NA, 2.764, 5.304, NA, NA~
## $ PLSO2CRT  <dbl> NA, NA, 3.225, 3.738, 3.767, NA, NA, 1.306, 0.053, NA, NA, 0~
## $ PLCO2CRT  <dbl> NA, NA, 1816.358, 2105.428, 2121.615, NA, NA, 1922.259, 1937~
## $ PLCH4CRT  <dbl> NA, NA, 0.074, 0.085, 0.086, NA, NA, 0.218, 0.037, NA, NA, 0~
## $ PLN2OCRT  <dbl> NA, NA, 0.015, 0.017, 0.017, NA, NA, 0.032, 0.004, NA, NA, 0~
## $ PLC2ECRT  <dbl> NA, NA, 1822.316, 2112.334, 2128.574, NA, NA, 1936.762, 1939~
## $ PLHGCRT   <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ UNNOX     <dbl> NA, NA, 22.889, 30.225, 41.920, NA, 2.385, 470.286, 123.742,~
## $ UNNOXOZ   <dbl> NA, NA, 8.668, 11.445, 15.872, NA, 0.903, 146.053, 58.418, N~
## $ UNSO2     <dbl> NA, NA, 1.998, 3.394, 3.799, NA, 1.174, 195.272, 1.236, NA, ~
## $ UNCO2     <dbl> NA, NA, 1125.234, 1911.729, 2139.649, NA, 386.918, 287402.04~
## $ UNCH4     <dbl> 0.000, NA, 91.133, 154.831, 173.290, NA, 31.337, 65170.670, ~
## $ UNN2O     <dbl> 0.000, NA, 18.227, 30.966, 34.658, NA, 6.267, 9479.370, 170.~
## $ UNCO2E    <dbl> 0.000, NA, 1128.925, 1918.000, 2146.667, NA, 388.187, 289570~
## $ UNHG      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ UNHTI     <dbl> 0, NA, 13779, 23410, 26201, NA, 4738, 2687351, 773377, NA, N~
## $ UNHTIOZ   <dbl> 0.000, NA, 5218.001, 8862.999, 9920.999, NA, 1791.000, 89815~
## $ UNHTIT    <dbl> NA, 71938, 13779, 23410, 26201, 28859, 4738, 2687351, 773377~
## $ UNHTIOZT  <dbl> NA, 61081.000, 5218.001, 8862.999, 9920.999, 0.000, 1791.000~
## $ UNNOXSRC  <chr> NA, NA, "EIA", "EIA", "EIA", NA, "EIA", NA, "EIA", NA, NA, "~
## $ UNNOZSRC  <chr> NA, NA, "EIA", "EIA", "EIA", NA, "EIA", "EIA", "EIA", NA, NA~
## $ UNSO2SRC  <chr> NA, NA, "EIA", "EIA", "EIA", NA, "EIA", NA, "EIA", NA, NA, "~
## $ UNCO2SRC  <chr> NA, NA, "EIA", "EIA", "EIA", NA, "EIA", "EIA", "EIA", NA, NA~
## $ UNCH4SRC  <chr> "EIA", NA, "EIA", "EIA", "EIA", NA, "EIA", "EIA", "EIA", NA,~
## $ UNN2OSRC  <chr> "EIA", NA, "EIA", "EIA", "EIA", NA, "EIA", "EIA", "EIA", NA,~
## $ UNC2ESRC  <chr> "Calculated", NA, "Calculated", "Calculated", "Calculated", ~
## $ UNHGSRC   <chr> "--", "--", "--", "--", "--", "--", "--", "--", "--", "--", ~
## $ UNHTISRC  <chr> NA, "EIA", "EIA", "EIA", "EIA", "EIA", "EIA", "EIA", "EIA", ~
## $ UNHOZSRC  <chr> NA, "EIA", "EIA", "EIA", "EIA", "EIA", "EIA", "EIA", "EIA", ~
## $ BIONOX    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ BIONOXOZ  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ BIOSO2    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ BIOCO2    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ BIOCH4    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ BION2O    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ BIOCO2E   <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ CHPCHTI   <dbl> 0, NA, NA, NA, NA, NA, NA, 1398970, NA, NA, NA, NA, NA, NA, ~
## $ CHPCHTIOZ <dbl> 0.0, NA, NA, NA, NA, NA, NA, 467559.3, NA, NA, NA, NA, NA, N~
## $ CHPNOX    <dbl> NA, NA, NA, NA, NA, NA, NA, 244.819, NA, NA, NA, NA, NA, NA,~
## $ CHPNOXOZ  <dbl> NA, NA, NA, NA, NA, NA, NA, 76.032, NA, NA, NA, NA, NA, NA, ~
## $ CHPSO2    <dbl> NA, NA, NA, NA, NA, NA, NA, 101.654, NA, NA, NA, NA, NA, NA,~
## $ CHPCO2    <dbl> NA, NA, NA, NA, NA, NA, NA, 149614.5, NA, NA, NA, NA, NA, NA~
## $ CHPCH4    <dbl> 0.00, NA, NA, NA, NA, NA, NA, 33926.26, NA, NA, NA, NA, NA, ~
## $ CHPN2O    <dbl> 0.000, NA, NA, NA, NA, NA, NA, 4934.729, NA, NA, NA, NA, NA,~
## $ CHPCO2E   <dbl> 0.0, NA, NA, NA, NA, NA, NA, 150743.3, NA, NA, NA, NA, NA, N~
## $ PLHTRT    <dbl> NA, NA, 11121.065, 12890.969, 12990.084, NA, -20076.271, 898~
## $ PLGENACL  <dbl> 0, 0, 0, 0, 0, 0, 0, 143360, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ PLGENAOL  <dbl> 0.000, 0.000, 1239.000, 1816.000, 2017.000, 0.000, -236.000,~
## $ PLGENAGS  <dbl> 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 46661.65, 0.~
## $ PLGENANC  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ PLGENAHY  <dbl> 0, 20961, 0, 0, 0, 8383, 0, 0, 0, 0, 42499, 0, 0, 0, 20403, ~
```

```
## $ PLGENABM  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ PLGENAWI  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ PLGENASO  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ PLGENAGT  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ PLGENAOF  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ PLGENAOP  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, -2443, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ PLGENATN  <dbl> 0, 0, 1239, 1816, 2017, 0, -236, 143360, 46662, 0, 0, 525, 0~
## $ PLGENATR  <dbl> 0, 20961, 0, 0, 0, 8383, 0, 0, 0, 0, 42499, 0, 0, 0, 20403, ~
## $ PLGENATO  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, -2443, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ PLGENATH  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ PLGENACY  <dbl> 0, 0, 1239, 1816, 2017, 0, -236, 143360, 46662, 0, 0, 525, 0~
## $ PLGENACN  <dbl> 0, 20961, 0, 0, 0, 8383, 0, 0, 0, 0, 42499, 0, 0, 0, 20403, ~
## $ PLGENACO  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, -2443, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ PLCLPR    <dbl> NA, 0, 0, 0, 0, 0, NA, 1, 0, NA, 0, 0, NA, 0, 0, 0, 0, 0, 0,~
## $ PLOLPR    <dbl> NA, 0.000000e+00, 1.000000e+00, 1.000000e+00, 1.000000e+00, ~
## $ PLGSPR    <dbl> NA, 0.0000000, 0.0000000, 0.0000000, 0.0000000, 0.0000000, N~
## $ PLNCPR    <dbl> NA, 0, 0, 0, 0, 0, NA, 0, 0, NA, 0, 0, NA, 0, 0, 0, 0, 0, 0,~
## $ PLHYPR    <dbl> NA, 1, 0, 0, 0, 1, NA, 0, 0, NA, 1, 0, NA, 0, 1, 1, 1, 0, 0,~
## $ PLBMPR    <dbl> NA, 0, 0, 0, 0, 0, NA, 0, 0, NA, 0, 0, NA, 0, 0, 0, 0, 0, 0,~
## $ PLWIPR    <dbl> NA, 0, 0, 0, 0, 0, NA, 0, 0, NA, 0, 0, NA, 0, 0, 0, 0, 0, 0,~
## $ PLSOPR    <dbl> NA, 0, 0, 0, 0, 0, NA, 0, 0, NA, 0, 0, NA, 0, 0, 0, 0, 0, 0,~
## $ PLGTPR    <dbl> NA, 0, 0, 0, 0, 0, NA, 0, 0, NA, 0, 0, NA, 0, 0, 0, 0, 0, 0,~
## $ PLOFPR    <dbl> NA, 0, 0, 0, 0, 0, NA, 0, 0, NA, 0, 0, NA, 0, 0, 0, 0, 0, 0,~
## $ PLOPPR    <dbl> NA, 0, 0, 0, 0, 0, NA, 0, 0, NA, 0, 0, NA, 0, 0, 0, 0, 0, 0,~
## $ PLTNPR    <dbl> NA, 0, 1, 1, 1, 0, NA, 1, 1, NA, 0, 1, NA, 1, 0, 0, 0, 1, 1,~
## $ PLTRPR    <dbl> NA, 1, 0, 0, 0, 1, NA, 0, 0, NA, 1, 0, NA, 0, 1, 1, 1, 0, 0,~
## $ PLTOPR    <dbl> NA, 0, 0, 0, 0, 0, NA, 0, 0, NA, 0, 0, NA, 0, 0, 0, 0, 0, 0,~
## $ PLTHPR    <dbl> NA, 0, 0, 0, 0, 0, NA, 0, 0, NA, 0, 0, NA, 0, 0, 0, 0, 0, 0,~
## $ PLCYPR    <dbl> NA, 0, 1, 1, 1, 0, NA, 1, 1, NA, 0, 1, NA, 1, 0, 0, 0, 1, 1,~
## $ PLCNPR    <dbl> NA, 1, 0, 0, 0, 1, NA, 0, 0, NA, 1, 0, NA, 0, 1, 1, 1, 0, 0,~
## $ PLCOPR    <dbl> NA, 0, 0, 0, 0, 0, NA, 0, 0, NA, 0, 0, NA, 0, 0, 0, 0, 0, 0,~
```

```r
# Process data -------------------------------------------------------------

# Filter plants based on:
# Southeastern states (plus Texas)
# Plants with Capacity Factor > 1% and Nameplate Capacity > 1 MW
# Exclude plants where county info is missing

southern_states <- c("FL", "GA", "AL", "MS", "LA", "AR", "TN", "NC", "SC",
                     "KY", "WV", "VA", "TX")
cf_limit <- 0.01
npcap_limit <- 1

plnt23 <- plnt23_raw %>%
  filter(PSTATABB %in% southern_states,
         NAMEPCAP >= npcap_limit,
         CAPFAC >= cf_limit,
         !is.na(FIPSCNTY))

counties_count <- plnt23 %>%
  group_by(PSTATABB, FIPSCNTY, CNTYNAME, PLFUELCT) %>%
  count() %>%
  ungroup() %>%
```

```r
  pivot_wider(id_cols = c("PSTATABB", "FIPSCNTY", "CNTYNAME"),
              names_from = "PLFUELCT",
              values_from = "n")

counties_mw <- plnt23 %>%
  group_by(PSTATABB, FIPSCNTY, CNTYNAME, PLFUELCT) %>%
  summarize(capacity = sum(NAMEPCAP)) %>%
  ungroup() %>%
  pivot_wider(id_cols = c("PSTATABB", "FIPSCNTY", "CNTYNAME"),
              names_from = "PLFUELCT",
              values_from = "capacity")

counties_count$allFF <- rowSums(counties_count[c("GAS", "COAL", "OIL", "OFSL")],
                                na.rm = TRUE)
counties_count$windsolar <- rowSums(counties_count[c("WIND", "SOLAR")],
                                    na.rm = TRUE)

counties_cat <- counties_count %>%
  mutate(Category = case_when(
    allFF > 0 & windsolar > 0 ~ "Contains FF and solar/wind",
    allFF > 0 ~ "Contains FF",
    windsolar > 0 ~ "Contains solar/wind",
    allFF == 0 & windsolar ==0 ~"Other plant"
  )) %>%
  mutate(FIPSTATE = case_when(
    PSTATABB == "FL" ~ "12",
    PSTATABB == "GA" ~ "13",
    PSTATABB == "AL" ~ "01",
    PSTATABB == "MS" ~ "28",
    PSTATABB == "LA" ~ "22",
    PSTATABB == "AR" ~ "05",
    PSTATABB == "TN" ~ "47",
    PSTATABB == "NC" ~ "37",
    PSTATABB == "SC" ~ "45",
    PSTATABB == "KY" ~ "21",
    PSTATABB == "WV" ~ "54",
    PSTATABB == "VA" ~ "51",
    PSTATABB == "TX" ~ "48",
    PSTATABB == "DC" ~ "11")) %>%
  mutate(FIPSCODE = paste0(FIPSTATE,FIPSCNTY)) %>%
  select(FIPSCODE,PSTATABB,CNTYNAME,Category)

#write.csv(counties_cat, "Data/Processed/counties_category.csv", row.names = FALSE)
```

Join

```r
CVI <- cvi_county_best_cols %>%
  rename(FIPSCODE = fips_county)

colnames(CVI)
```

```
## [1] "state"
```

```
## [2] "FIPSCODE"
## [3] "county_name"
## [4] "current_adult_asthma"
## [5] "below_poverty"
## [6] "no_high_school_diploma"
## [7] "riverine_flooding_annualized_frequency_log"
## [8] "population_log"
```

```
county_cat <- counties_cat

cvi_egrid_join <- left_join(CVI,county_cat,by = "FIPSCODE") %>%
  mutate(Category = replace_na(Category, "No plant")) %>%
  select("state","FIPSCODE","Category","population_log","current_adult_asthma",
         "below_poverty","no_high_school_diploma",
         "riverine_flooding_annualized_frequency_log") %>%
  filter(Category != "Other plant")

cvi_egrid_join
```

```
## # A tibble: 1,191 x 8
##    state FIPSCODE Category     population_log current_adult_asthma below_poverty
##    <chr> <chr>    <chr>                 <dbl>                <dbl>         <dbl>
##  1 AL    01001    No plant              11.0                  10.8         0.596
##  2 AL    01003    No plant              12.4                  10.0         0.500
##  3 AL    01005    No plant              10.1                  11.7         0.856
##  4 AL    01007    No plant              10.0                  10.1         0.126
##  5 AL    01009    No plant              11.0                  10.4         0.496
##  6 AL    01011    No plant               9.22                 12.8         0.598
##  7 AL    01013    No plant               9.83                 11.5         0.760
##  8 AL    01015    Contains so~          11.7                  11.6         0.689
##  9 AL    01017    Contains so~          10.4                  11.0         0.600
## 10 AL    01027    No plant               9.56                 11.1         0.615
## # i 1,181 more rows
## # i 2 more variables: no_high_school_diploma <dbl>,
## #   riverine_flooding_annualized_frequency_log <dbl>
```

```
#write.csv(cvi_egrid_join, "Data/Processed/cvi_egrid_join.csv", row.names = FALSE)
```

Figures and Data Exploration

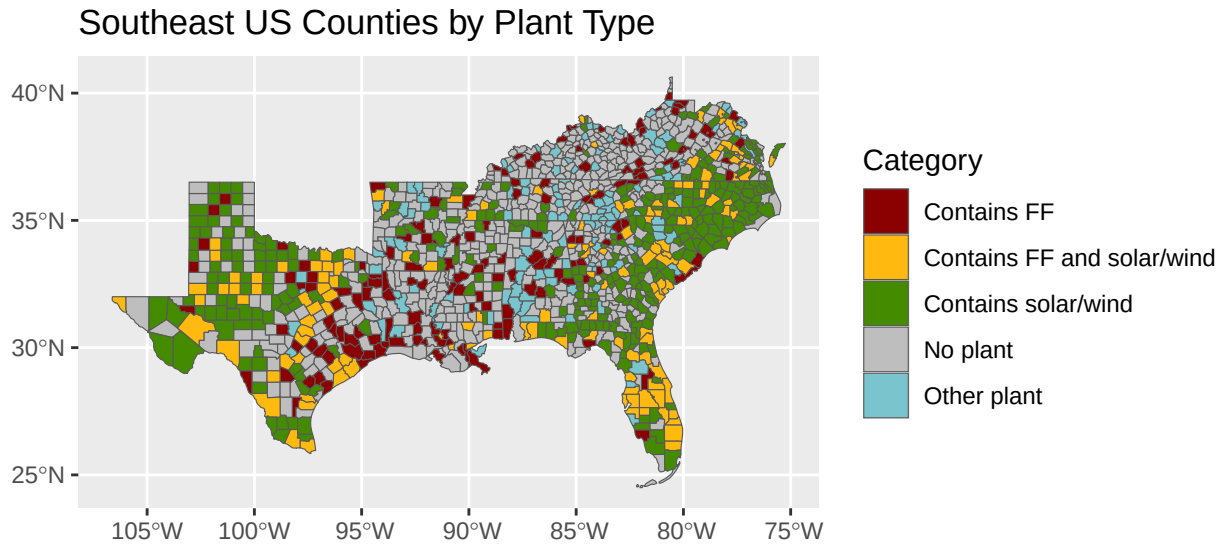
Plant Categories Map

```
southern_fps <- c(
  "12","13","01","28","22","05","47","37","45","21","54","51","48")

SE_Counties <- st_read(here('Data/Raw/counties/cb_2018_us_county_20m.shp')) %>%
  filter(STATEFP %in% southern_fps) %>%
  mutate(FIPSCODE = as.character(GEOID))
```

```
## Reading layer ‘cb_2018_us_county_20m’ from data source
```

```
##   '/home/guest/ENV710/ENV710_GroupProj/Data/Raw/counties/cb_2018_us_county_20m.shp'
##   using driver 'ESRI Shapefile'
## Simple feature collection with 3220 features and 9 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -179.1743 ymin: 17.91377 xmax: 179.7739 ymax: 71.35256
## Geodetic CRS:  NAD83
```

```
counties_category <- counties_cat


map_countycat <- left_join(SE_Counties, counties_category, by = "FIPSCODE") %>%
  mutate(Category = replace_when(x = Category, is.na(Category) ~ "No plant" ))

map_cat<-ggplot()+
  geom_sf(data = map_countycat, aes(fill = Category))+
  ggtitle("Southeast US Counties by Plant Type")+
  scale_fill_manual(
    values = c("Contains FF" = "darkred",
               "Contains FF and solar/wind" = "darkgoldenrod1",
               "Contains solar/wind" = "chartreuse4",
               "No plant" = "gray",
               "Other plant" = "cadetblue3"))+
  labs(caption = "Counties in the Southeastern United States categorized by the presence of
  power plants within county boundaries according to 2023 eGRID data.")

map_cat
```



Counties in the Southeastern United States categorized by the presence of power plants within county boundaries according to 2023 eGRID data.

```
#ggsave(map_cat,
 #      filename = "Scripts/Analysis/Figures/map_categories.jpg",
  #     height = 10,
   #    width = 20,
    #   units = "cm")
```

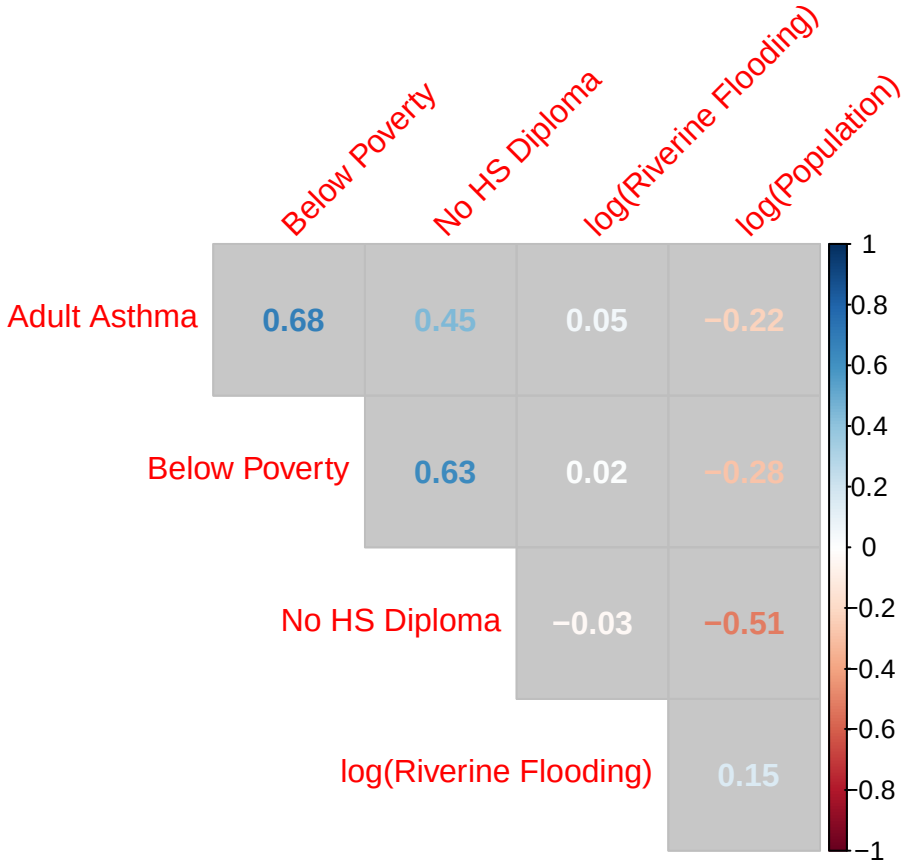
Correlation Plot

```
color_levels <- c("Contains FF" = "darkred",
                  "Contains FF and solar/wind" = "darkgoldenrod1",
                  "Contains solar/wind" = "chartreuse4",
                  "No Plant" = "grey")

# Load Data ----------------------------------------------------------------

counties_category <- counties_cat %>%
  clean_names()

cvi_by_county <- cvi_county_best_cols

cvi_egrid_df <- left_join(
  cvi_by_county,
  counties_category %>% select(fipscode, category),
  by = c("fips_county" = "fipscode")
```

```r
) %>%
  mutate(category = ifelse(is.na(category), "No Plant", category)) %>%
  filter(!str_detect(category, "Other")) %>%
  mutate(category = ordered(category))

summary(cvi_egrid_df)
```

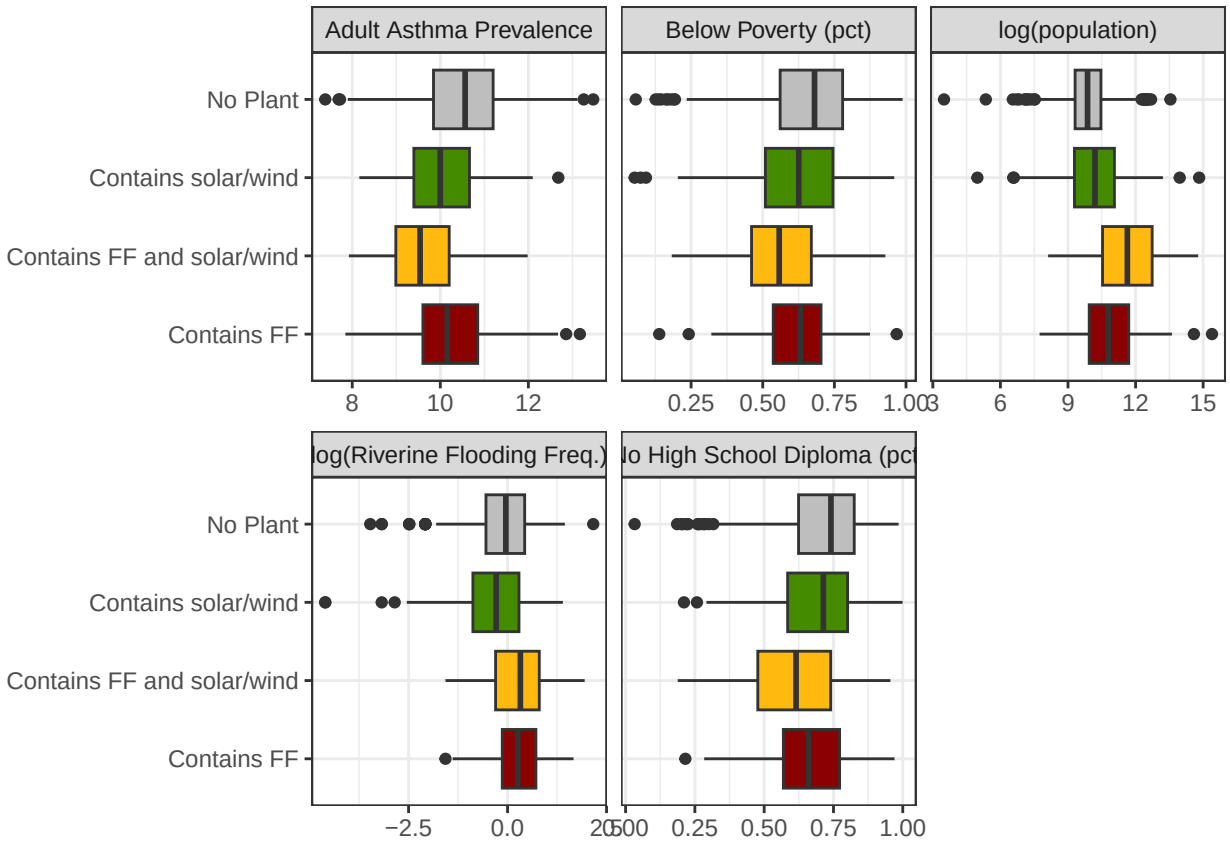
```
##     state           fips_county        county_name        current_adult_asthma
##  Length:1191        Length:1191        Length:1191        Min.   : 7.387
##  Class :character   Class :character   Class :character   1st Qu.: 9.523
##  Mode  :character   Mode  :character   Mode  :character   Median :10.205
##                                                           Mean   :10.268
##                                                           3rd Qu.:10.951
##                                                           Max.   :13.473
##  below_poverty    no_high_school_diploma
##  Min.   :0.0530   Min.   :0.03265
##  1st Qu.:0.5250   1st Qu.:0.58077
##  Median :0.6449   Median :0.71344
##  Mean   :0.6318   Mean   :0.68457
##  3rd Qu.:0.7530   3rd Qu.:0.80315
##  Max.   :0.9881   Max.   :0.99940
##  riverine_flooding_annualized_frequency_log population_log
##  Min.   :    -Inf                           Min.   : 3.497
##  1st Qu.:-0.61310                           1st Qu.: 9.456
##  Median :-0.02405                           Median :10.144
##  Mean   :    -Inf                           Mean   :10.305
##  3rd Qu.: 0.48551                           3rd Qu.:11.091
##  Max.   : 2.16428                           Max.   :15.392
##                        category
##  Contains FF               :151
##  Contains FF and solar/wind:140
##  Contains solar/wind       :319
##  No Plant                  :581
##
##
```

```r
# Correlation between predictor vars -------------------------------------

predictor_vars <- c("Adult Asthma"="current_adult_asthma",
                    "Below Poverty"="below_poverty",
                    "No HS Diploma"="no_high_school_diploma",
                    "log(Riverine Flooding)"="riverine_flooding_annualized_frequency_log",
                    "log(Population)"="population_log")

# check correlation
corr_plot <- cvi_egrid_df %>%
  mutate(riverine_flooding_annualized_frequency_log = case_when(
    riverine_flooding_annualized_frequency_log==-Inf ~ -1000000,
    TRUE ~ riverine_flooding_annualized_frequency_log)) %>%
  select(c(current_adult_asthma:population_log)) %>%
  rename(all_of(predictor_vars)) %>%
  cor() %>%
  corrplot(method = "number", type = "upper", bg="#c7c7c7", diag=FALSE, tl.srt = 45)
```



Boxplots

```
df_to_plot <- cvi_egrid_df %>%
  pivot_longer(cols = c(current_adult_asthma:population_log),
               names_to = "metric") %>%
  mutate(metric = case_when(metric == "below_poverty" ~ "Below Poverty (pct)",
                            metric == "current_adult_asthma" ~ "Adult Asthma Prevalence",
                            metric == "no_high_school_diploma" ~ "No High School Diploma (pct)",
                            metric == "riverine_flooding_annualized_frequency_log" ~ "log(Riverine Flooc
                            metric == "population_log" ~ "log(population)")) %>%
  filter(value != -Inf)

cvi_boxplots <- ggplot(df_to_plot, aes(x = value, y = category, fill = category)) +
  geom_boxplot() +
  facet_wrap(vars(metric),
             scales = "free_x") +
  scale_fill_manual(values = color_levels) +
  theme_bw() +
  theme(legend.position = "None") +
  labs(x = NULL, y = NULL)

cvi_boxplots
```



Analysis

ANOVA

```
cvi_egrid_join <- cvi_egrid_join
riverine_notinfinity <- cvi_egrid_join %>%
  filter(is.finite(riverine_flooding_annualized_frequency_log))

unique(cvi_egrid_join$Category)
```

```
## [1] "No plant"                    "Contains solar/wind"
## [3] "Contains FF and solar/wind" "Contains FF"
```

```
windsolar <-
  cvi_egrid_join %>%
  filter(Category == "Contains solar/wind")

solarwindff  <-
  cvi_egrid_join %>%
  filter(Category == "Contains FF and solar/wind")

ff <-
```

```r
  cvi_egrid_join %>%
  filter(Category == "Contains FF")

noplant<-
  cvi_egrid_join %>%
  filter(Category == "No plant")

noplantflood <- riverine_notinfinity %>%
  filter(Category == "No plant")

windsolarflood <- riverine_notinfinity %>%
  filter(Category == "Contains solar/wind")

windsolarflood_summary <- data.frame(
  mean = sapply(windsolarflood[sapply(windsolarflood, is.numeric)], mean, na.rm = TRUE),
  sd   = sapply(windsolarflood[sapply(windsolarflood, is.numeric)], sd, na.rm = TRUE)
)

noplantflood_summary <- data.frame(
  mean = sapply(noplantflood[sapply(noplantflood, is.numeric)], mean, na.rm = TRUE),
  sd   = sapply(noplantflood[sapply(noplantflood, is.numeric)], sd, na.rm = TRUE)
)

windsolar_summary <- data.frame(
  mean = sapply(windsolar[sapply(windsolar, is.numeric)], mean, na.rm = TRUE),
  sd   = sapply(windsolar[sapply(windsolar, is.numeric)], sd, na.rm = TRUE)
)

windsolarff_summary <- data.frame(
  mean = sapply(solarwindff[sapply(solarwindff, is.numeric)], mean, na.rm = TRUE),
  sd   = sapply(solarwindff[sapply(solarwindff, is.numeric)], sd, na.rm = TRUE)
)


ff_summary <- data.frame(
  mean = sapply(ff[sapply(ff, is.numeric)], mean, na.rm = TRUE),
  sd   = sapply(ff[sapply(ff, is.numeric)], sd, na.rm = TRUE)
)

noplant_summary <- data.frame(
  mean = sapply(noplant[sapply(noplant, is.numeric)], mean, na.rm = TRUE),
  sd   = sapply(noplant[sapply(noplant, is.numeric)], sd, na.rm = TRUE)
)

colnames(cvi_egrid_join)
```

```
## [1] "state"
## [2] "FIPSCODE"
## [3] "Category"
## [4] "population_log"
## [5] "current_adult_asthma"
## [6] "below_poverty"
## [7] "no_high_school_diploma"
```

```
## [8] "riverine_flooding_annualized_frequency_log"
```

```r
#ANOVA

#asthma
asthma_ANOVA <- aov(current_adult_asthma ~ Category, data = cvi_egrid_join)
summary(asthma_ANOVA)
```

```
##               Df Sum Sq Mean Sq F value Pr(>F)
## Category       3  109.8   36.59   39.76 <2e-16 ***
## Residuals   1187 1092.5    0.92
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
TukeyHSD(asthma_ANOVA)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = current_adult_asthma ~ Category, data = cvi_egrid_join)
##
## $Category
##                                                      diff         lwr
## Contains FF and solar/wind-Contains FF         -0.5985927 -0.88817597
## Contains solar/wind-Contains FF                -0.2072290 -0.45103507
## No plant-Contains FF                            0.2815878  0.05613351
## Contains solar/wind-Contains FF and solar/wind  0.3913637  0.14114121
## No plant-Contains FF and solar/wind             0.8801805  0.64780250
## No plant-Contains solar/wind                    0.4888167  0.31682117
##                                                       upr     p adj
## Contains FF and solar/wind-Contains FF         -0.3090095 0.0000007
## Contains solar/wind-Contains FF                 0.0365771 0.1274826
## No plant-Contains FF                            0.5070420 0.0073611
## Contains solar/wind-Contains FF and solar/wind  0.6415863 0.0003542
## No plant-Contains FF and solar/wind             1.1125585 0.0000000
## No plant-Contains solar/wind                    0.6608123 0.0000000
```

```r
#population
pop_ANOVA <- aov(population_log ~ Category, data = cvi_egrid_join)
summary(pop_ANOVA)
```

```
##               Df Sum Sq Mean Sq F value Pr(>F)
## Category       3  407.7  135.91   87.68 <2e-16 ***
## Residuals   1187 1839.9    1.55
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
TukeyHSD(pop_ANOVA)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
```

```
##
## Fit: aov(formula = population_log ~ Category, data = cvi_egrid_join)
##
## $Category
##                                                      diff        lwr
## Contains FF and solar/wind-Contains FF          0.7760576  0.4002586
## Contains solar/wind-Contains FF                -0.7113321 -1.0277250
## No plant-Contains FF                           -0.9822279 -1.2748051
## Contains solar/wind-Contains FF and solar/wind -1.4873896 -1.8121093
## No plant-Contains FF and solar/wind            -1.7582854 -2.0598478
## No plant-Contains solar/wind                   -0.2708958 -0.4940985
##                                                        upr     p adj
## Contains FF and solar/wind-Contains FF          1.15185655 0.0000008
## Contains solar/wind-Contains FF                -0.39493920 0.0000001
## No plant-Contains FF                           -0.68965057 0.0000000
## Contains solar/wind-Contains FF and solar/wind -1.16266999 0.0000000
## No plant-Contains FF and solar/wind            -1.45672305 0.0000000
## No plant-Contains solar/wind                   -0.04769307 0.0099192
```

```r
#poverty
poverty_ANOVA <- aov(below_poverty ~ Category, data = cvi_egrid_join)
summary(poverty_ANOVA)
```

```
##               Df Sum Sq Mean Sq F value   Pr(>F)
## Category       3   1.40  0.4682   17.27 5.59e-11 ***
## Residuals   1187  32.18  0.0271
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
TukeyHSD(poverty_ANOVA)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = below_poverty ~ Category, data = cvi_egrid_join)
##
## $Category
##                                                        diff          lwr
## Contains FF and solar/wind-Contains FF         -0.065047966 -0.114748162
## Contains solar/wind-Contains FF                 0.003405328 -0.038438288
## No plant-Contains FF                            0.042269251  0.003575298
## Contains solar/wind-Contains FF and solar/wind  0.068453294  0.025508445
## No plant-Contains FF and solar/wind             0.107317217  0.067434967
## No plant-Contains solar/wind                    0.038863923  0.009344902
##                                                        upr     p adj
## Contains FF and solar/wind-Contains FF         -0.01534777 0.0043544
## Contains solar/wind-Contains FF                 0.04524894 0.9967553
## No plant-Contains FF                            0.08096320 0.0258588
## Contains solar/wind-Contains FF and solar/wind  0.11139814 0.0002567
## No plant-Contains FF and solar/wind             0.14719947 0.0000000
## No plant-Contains solar/wind                    0.06838294 0.0040603
```

```r
#hs diploma
hs_ANOVA <- aov(no_high_school_diploma ~ Category, data = cvi_egrid_join)
summary(hs_ANOVA)
```

```
##               Df Sum Sq Mean Sq F value   Pr(>F)
## Category       3  1.358  0.4527   18.19 1.52e-11 ***
## Residuals   1187 29.542  0.0249
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
TukeyHSD(hs_ANOVA)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = no_high_school_diploma ~ Category, data = cvi_egrid_join)
##
## $Category
##                                                        diff          lwr
## Contains FF and solar/wind-Contains FF         -0.05152662 -0.099145065
## Contains solar/wind-Contains FF                 0.03230005 -0.007790905
## No plant-Contains FF                            0.05249091  0.015417698
## Contains solar/wind-Contains FF and solar/wind  0.08382666  0.042680604
## No plant-Contains FF and solar/wind             0.10401753  0.065805790
## No plant-Contains solar/wind                    0.02019087 -0.008091718
##                                                         upr     p adj
## Contains FF and solar/wind-Contains FF         -0.003908166 0.0279220
## Contains solar/wind-Contains FF                 0.072390997 0.1626567
## No plant-Contains FF                            0.089564127 0.0016020
## Contains solar/wind-Contains FF and solar/wind  0.124972719 0.0000011
## No plant-Contains FF and solar/wind             0.142229267 0.0000000
## No plant-Contains solar/wind                    0.048473451 0.2566263
```

```r
#riverine_flooding_annualized_frequency
riverine_ANOVA <- aov(riverine_flooding_annualized_frequency_log ~ Category,
                      data = riverine_notinfinity)
summary(riverine_ANOVA)
```

```
##               Df Sum Sq Mean Sq F value Pr(>F)
## Category       3   52.8  17.598   27.66 <2e-16 ***
## Residuals   1183  752.8   0.636
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
TukeyHSD(riverine_ANOVA)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = riverine_flooding_annualized_frequency_log ~ Category, data = riverine_notinfinit
##
```

```
## $Category
##                                                        diff         lwr
## Contains FF and solar/wind-Contains FF         -0.02576657 -0.26655214
## Contains solar/wind-Contains FF                -0.59410414 -0.79703182
## No plant-Contains FF                           -0.38455189 -0.57208162
## Contains solar/wind-Contains FF and solar/wind -0.56833757 -0.77659520
## No plant-Contains FF and solar/wind            -0.35878532 -0.55207007
## No plant-Contains solar/wind                    0.20955226  0.06616137
##                                                        upr     p adj
## Contains FF and solar/wind-Contains FF          0.2150190 0.9927108
## Contains solar/wind-Contains FF                -0.3911765 0.0000000
## No plant-Contains FF                           -0.1970222 0.0000009
## Contains solar/wind-Contains FF and solar/wind -0.3600800 0.0000000
## No plant-Contains FF and solar/wind            -0.1655006 0.0000120
## No plant-Contains solar/wind                    0.3529431 0.0010232
```

Model

```
data <- cvi_egrid_join

data1 <- data %>%
  filter(riverine_flooding_annualized_frequency_log != -Inf) %>%
  mutate(category_GLM = case_when(Category == "Contains FF" ~ 1,
                                  Category == "Contains solar/wind" ~ 1,
                                  Category == "Contains FF and solar/wind" ~ 1,
                                  T ~ 0 )) #Plant exists = 1; No Plant = 0



#-Logistic Regressions-------------
CategoryGLM1 <- glm(category_GLM ~ population_log,
                 data = data1,
                 family = "binomial")

CategoryGLM2 <- glm(category_GLM ~ current_adult_asthma,
                    data = data1,
                    family = "binomial")

CategoryGLM3 <- glm(category_GLM ~ below_poverty,
                    data = data1,
                    family = "binomial")

CategoryGLM4 <- glm(category_GLM ~ no_high_school_diploma,
                    data = data1,
                    family = "binomial")

CategoryGLM5 <- glm(category_GLM ~ riverine_flooding_annualized_frequency_log,
                    data = data1,
                    family = "binomial")

level1AIC <- as.data.frame(rbind(AIC(CategoryGLM1),
                                 AIC(CategoryGLM2),
```

```r
                                AIC(CategoryGLM3),
                                AIC(CategoryGLM4),
                                AIC(CategoryGLM5)))

# best performing model is GLM2 - using this to guide how
# we add a layer of complexity

CategoryGLM21 <- glm(category_GLM ~ current_adult_asthma+ population_log,
                     data = data1,
                     family = "binomial")

CategoryGLM23 <- glm(category_GLM ~ current_adult_asthma + below_poverty,
                     data = data1,
                     family = "binomial")
CategoryGLM24 <- glm(category_GLM ~ current_adult_asthma + no_high_school_diploma,
                      data = data1,
                      family = "binomial")
CategoryGLM25 <- glm(category_GLM ~ current_adult_asthma + riverine_flooding_annualized_frequency_log,
                      data = data1,
                      family = "binomial")


level2AIC <- as.data.frame(rbind(AIC(CategoryGLM21),
                                AIC(CategoryGLM23),
                                AIC(CategoryGLM24),
                                AIC(CategoryGLM25)))
min(level2AIC) # GLM25 (current_adult_asthma + riverine_flooding_annualized_frequency_log) has best AIC
```

```
## [1] 1492.096
```

```r
# using this to narrow choices

# level 3 complexity
CategoryGLM251 <- glm(category_GLM ~ current_adult_asthma + riverine_flooding_annualized_frequency_log +
                       data = data1,
                       family = "binomial")
CategoryGLM253 <- glm(category_GLM ~ current_adult_asthma + riverine_flooding_annualized_frequency_log +
                       data = data1,
                       family = "binomial")
CategoryGLM254 <- glm(category_GLM ~ current_adult_asthma + riverine_flooding_annualized_frequency_log
                       + no_high_school_diploma,
                       data = data1,
                       family = "binomial")

level3AIC <- as.data.frame(rbind(AIC(CategoryGLM251),
                                AIC(CategoryGLM253),
                                AIC(CategoryGLM254)))
min(level3AIC) # an improvement to add population_log
```

```
## [1] 1486.924
```

```r
# level 4 complexity
CategoryGLM2513 <- glm(category_GLM ~ current_adult_asthma + riverine_flooding_annualized_frequency_log
                         population_log + below_poverty,
                       data = data1,
                       family = "binomial")
CategoryGLM2514 <- glm(category_GLM ~ current_adult_asthma + riverine_flooding_annualized_frequency_log
                         population_log + no_high_school_diploma,
                       data = data1,
                       family = "binomial")

level4AIC <- as.data.frame(rbind(AIC(CategoryGLM2513),
                                 AIC(CategoryGLM2514)))
min(level4AIC) # it is only fractional improvement to add no_high_school_diploma
```

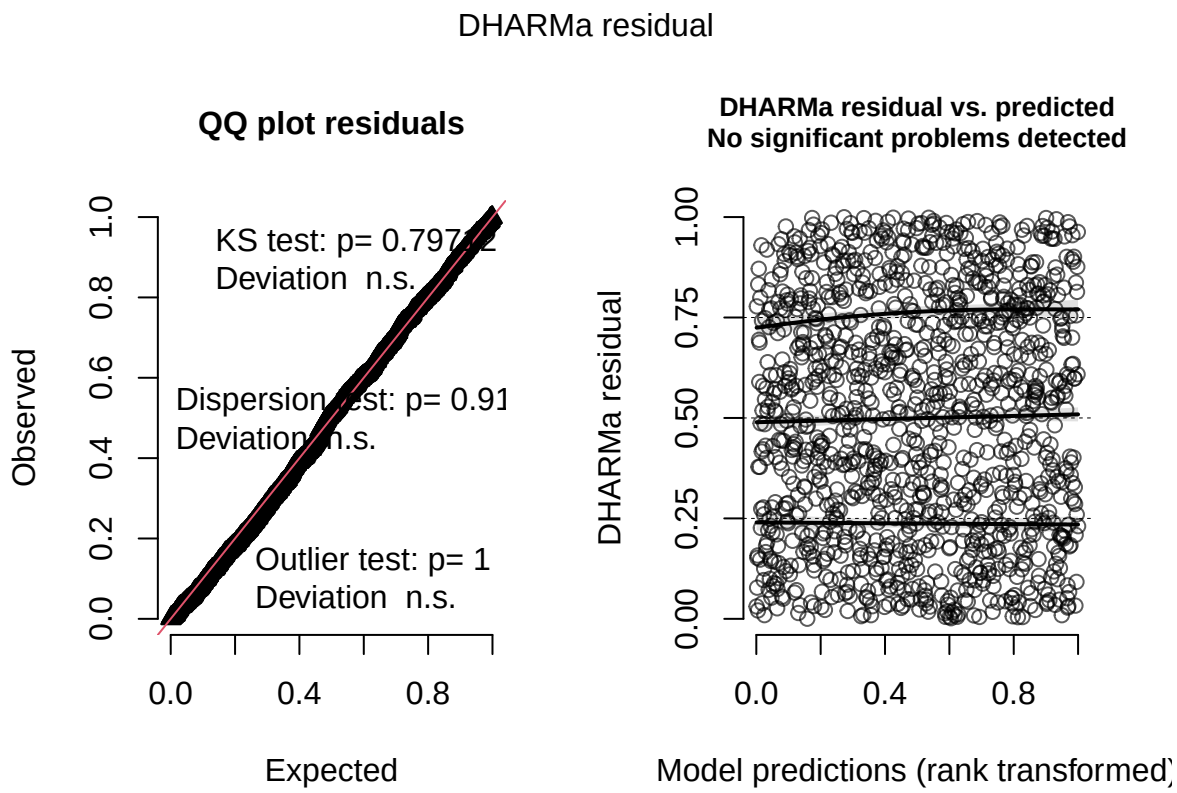
```
## [1] 1478.107
```

```r
# We decide this is not sufficient improvement to warrant additional complexity

# test if all explanatory variables together improve model any further:
# CategoryGLM25143 <- glm(category_GLM ~ current_adult_asthma + riverine_flooding_annualized_frequency_
#                          population_log + no_high_school_diploma + below_poverty,
#                        data = data1,
#                        family = "binomial")
# level5AIC <- as.data.frame(rbind(AIC(CategoryGLM25143)))
# level5AIC < min(level4AIC) # RETURNS FALSE


summary(CategoryGLM251)
```

```
##
## Call:
## glm(formula = category_GLM ~ current_adult_asthma + riverine_flooding_annualized_frequency_log +
##     population_log, family = "binomial", data = data1)
##
## Coefficients:
##                                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)                              -0.17916    0.94616  -0.189  0.84981
## current_adult_asthma                     -0.46175    0.06679  -6.914 4.73e-12
## riverine_flooding_annualized_frequency_log -0.22426    0.08408  -2.667  0.00765
## population_log                            0.48195    0.05695   8.462  < 2e-16
##
## (Intercept)
## current_adult_asthma                     ***
## riverine_flooding_annualized_frequency_log **
## population_log                           ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1644.8  on 1186  degrees of freedom
## Residual deviance: 1478.9  on 1183  degrees of freedom
## AIC: 1486.9
```

```
##
## Number of Fisher Scoring iterations: 4
```

```r
simulateResiduals(CategoryGLM251) %>% plot()
```




```r
# final model: category_GLM ~ current_adult_asthma + riverine_flooding_annualized_frequency_log + popul
```

Model visualization

```r
#filter out riverine infinity
riverine_notinfinity <- cvi_egrid_join %>%
  filter(is.finite(riverine_flooding_annualized_frequency_log))

riverine_vector <- c(rep(round(mean(
  riverine_notinfinity$riverine_flooding_annualized_frequency_log),2), 30))
population_vector <- c(rep(round(mean(cvi_egrid_join$population_log),2),30))
asthma_vector <- c(rep(round(mean(cvi_egrid_join$current_adult_asthma),2),30))
```

```r
summary(cvi_egrid_join$current_adult_asthma)
```

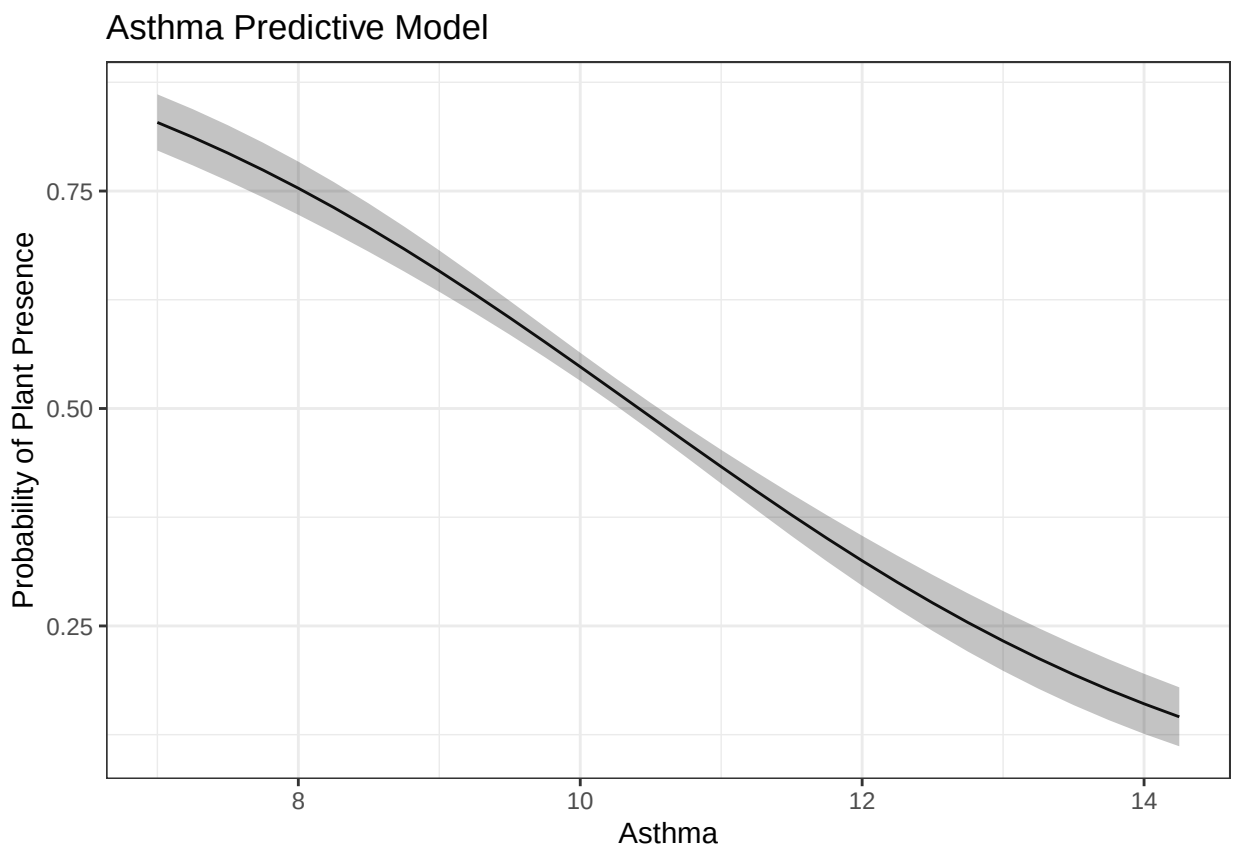
adult asthma visualization

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   7.387   9.523  10.205  10.268  10.951  13.473
```

```r
# Simulate asthma
asthma_sim <- seq(from = 7, to = 14.25, by = 0.25)
length(asthma_sim)
```

```
## [1] 30
```

```r
data_pred_asthma <- data.frame(asthma_sim, population_vector, riverine_vector)

colnames(data_pred_asthma) <- c("current_adult_asthma","population_log", "riverine_flooding_annualized_

# Use the original model to predict outcomes.
prediction_asthma <- predict(CategoryGLM251, newdata = data_pred_asthma, type = "response",
                        se.fit = TRUE)
# Pull out predictions.
data_fig_asthma <- data.frame(data_pred_asthma, prediction_asthma$fit,prediction_asthma$se.fit)

colnames(data_fig_asthma) <- c("asthma","population", "riverine", "probability", "se")

#plot
asthma_pred <- ggplot(data_fig_asthma, aes(x = asthma, y = probability)) +
  geom_line() +
  geom_ribbon(aes(ymin = probability - se,
                  ymax = probability + se), alpha = 0.3) +
  labs(x = "Asthma", y = "Probability of Plant Presence")+
  theme_bw()+
  ggtitle("Asthma Predictive Model")

asthma_pred
```




```
# Simulate population
summary(cvi_egrid_join$population_log)
```

population visualization

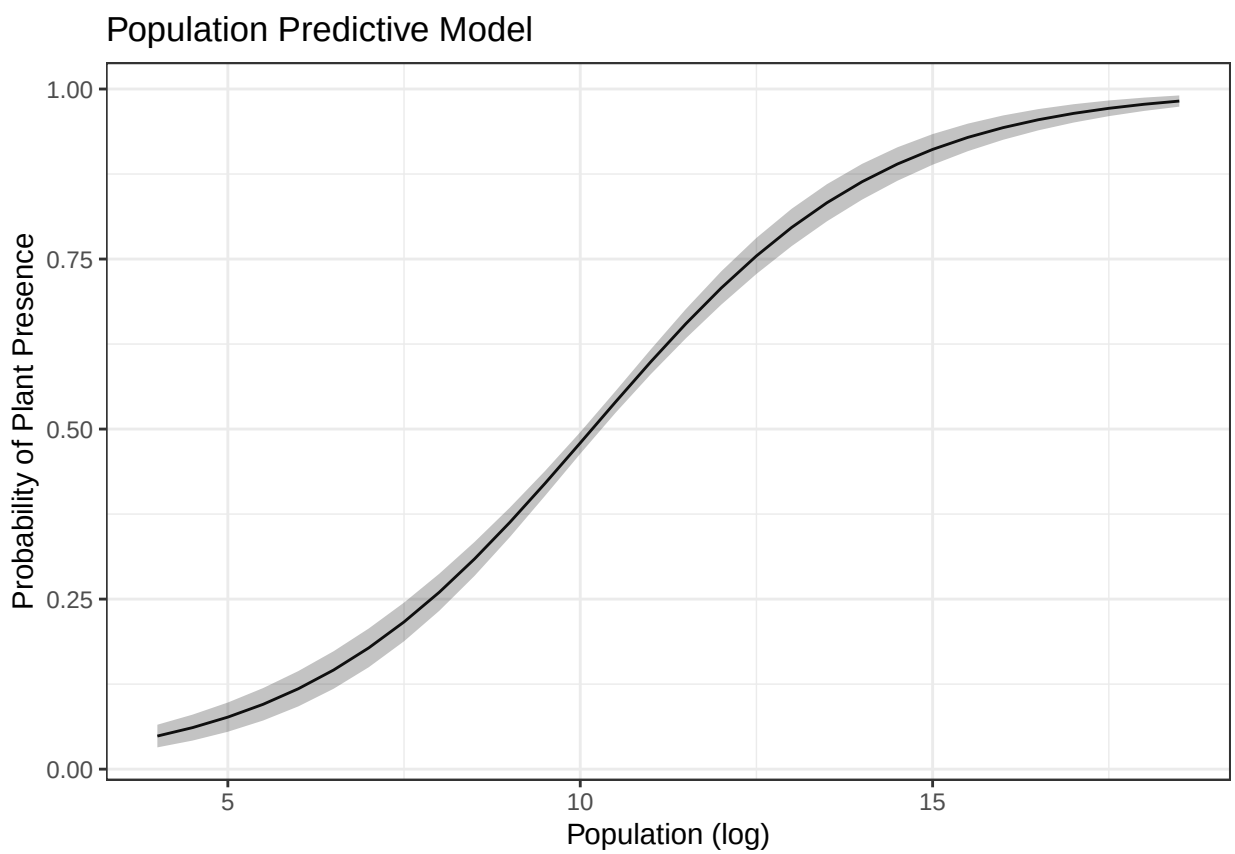
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   3.497   9.456  10.144  10.305  11.091  15.392
```

```
pop_sim <- seq(from = 4, to = 18.5, by = 0.5)
length(pop_sim)
```

```
## [1] 30
```

```
data_pred_pop <- data.frame(asthma_vector, pop_sim, riverine_vector)

colnames(data_pred_pop) <- c("current_adult_asthma","population_log", "riverine_flooding_annualized_fre

# Use the original model to predict outcomes.
prediction_population <- predict(CategoryGLM251, newdata = data_pred_pop, type = "response",
                         se.fit = TRUE)
# Pull out predictions.
```

```r
data_fig_pop <- data.frame(data_pred_pop, prediction_population$fit,prediction_population$se.fit)

colnames(data_fig_pop) <- c("asthma","population", "riverine","probability", "se")

#plot
pop_pred <- ggplot(data_fig_pop, aes(x = population, y = probability)) +
  geom_line() +
  geom_ribbon(aes(ymin = probability - se,
                  ymax = probability + se), alpha = 0.3) +
  labs(x = "Population (log)", y = "Probability of Plant Presence")+
  theme_bw()+
  ggtitle("Population Predictive Model")

pop_pred
```




```r
# Simulate riverine
summary(riverine_notinfinity$riverine_flooding_annualized_frequency_log)
```

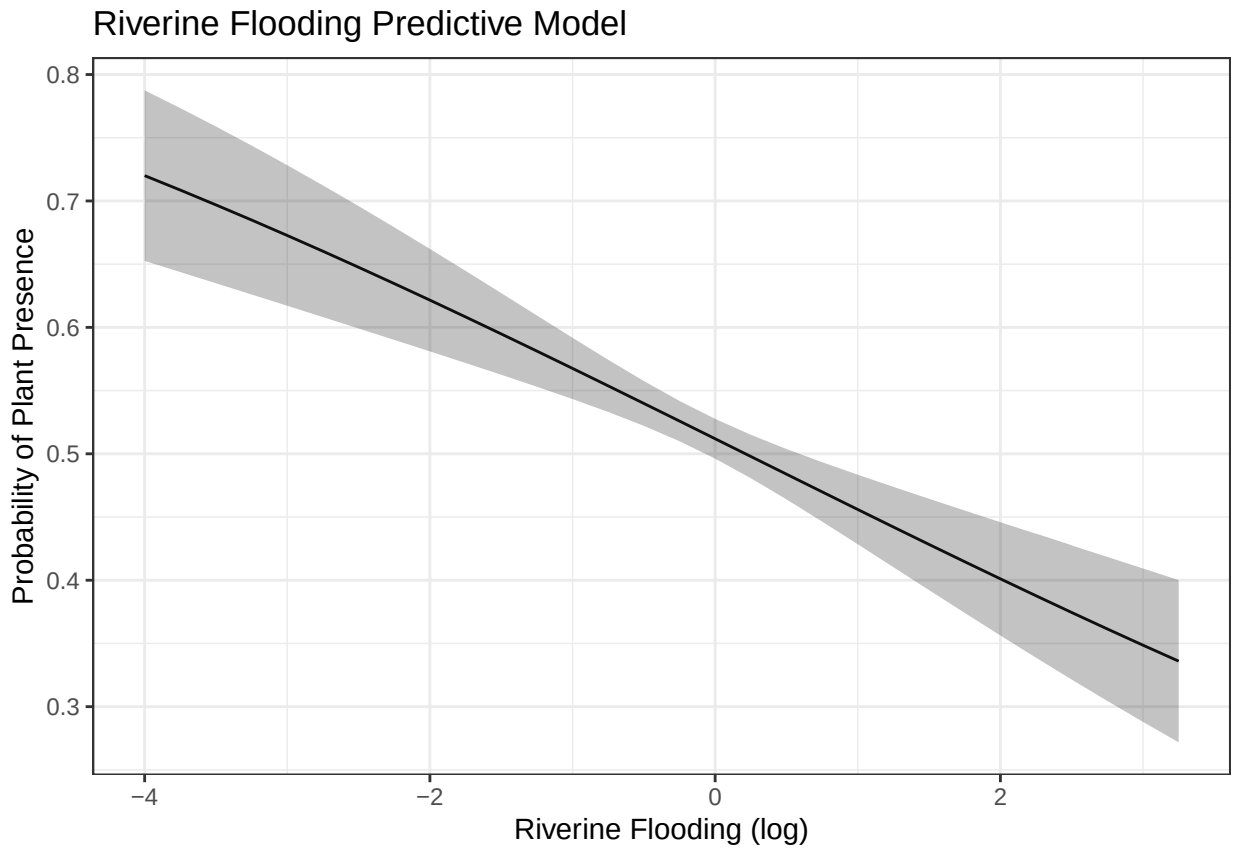
Riverine visualization

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## -4.60517 -0.61310 -0.01429 -0.09027  0.48551  2.16428
```

```r
riverine_sim <- seq(from = -4, to = 3.25, by = 0.25)
length(riverine_sim)
```

```
## [1] 30
```

```r
data_pred_riverine <- data.frame(asthma_vector, population_vector, riverine_sim)

colnames(data_pred_riverine) <- c("current_adult_asthma","population_log", "riverine_flooding_annualized

# Use the original model to predict outcomes.
prediction_riverine <- predict(CategoryGLM251, newdata = data_pred_riverine, type = "response",
                        se.fit = TRUE)
# Pull out predictions.
data_fig_riverine <- data.frame(data_pred_riverine, prediction_riverine$fit,prediction_riverine$se.fit)

colnames(data_fig_riverine) <- c("asthma","population", "riverine","probability", "se")

#plot
riverine_pred <- ggplot(data_fig_riverine, aes(x = riverine, y = probability)) +
  geom_line() +
  geom_ribbon(aes(ymin = probability - se,
                  ymax = probability + se), alpha = 0.3) +
  labs(x = "Riverine Flooding (log)", y = "Probability of Plant Presence")+
  theme_bw()+
  ggtitle("Riverine Flooding Predictive Model")

riverine_pred
```




```
grid<- plot_grid(pop_pred,riverine_pred,asthma_pred)

ggsave(grid,
       filename = "Scripts/Analysis/Figures/model_vis.jpg",
       height = 15,
       width = 30,
       units = "cm")
```